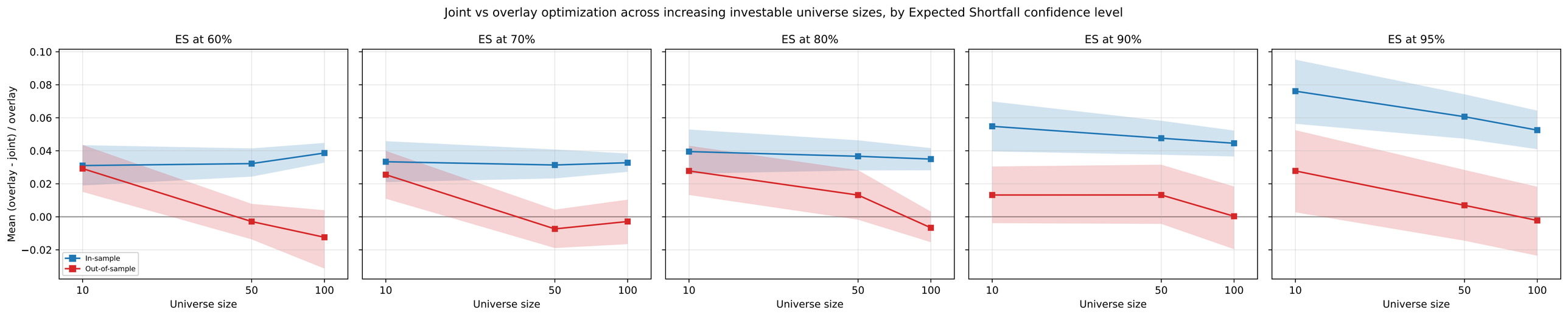
Joint vs overlay optimization across increasing investable universe sizes, by Expected Shortfall confidence level
ES at 60%
ES at 70%
ES at 80%
ES at 90%
ES at 95%
Mean (overlay - joint) / overlay
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
In-sample
Out-of-sample
10
50
100
Universe size